Late Refill Risk Prediction

Data-AI-Hackathon – Task 1, Challenge A

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Team Submission

March 31, 2026

Time-based split + patient holdout

1. Problem and grouping challenge

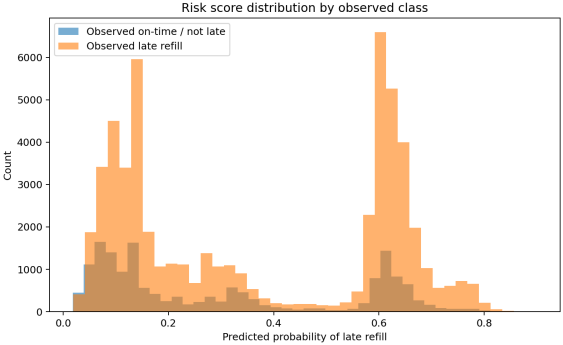
Goal

Predict **late refill** risk from historical pharmacy claims.

Main data challenge

- `PROD_SRVC_ID` is highly specific
- overly narrow grouping fragments refill chains
- tested **NDC11, NDC8, NDC7**

Best grouping
NDC7



Statistical Method for Refill Chains and Group Selection

Chain construction

- Consecutive fills within the same group were linked into a **refill chain**.
- For a fill at time t_i with days supply d_i, the expected refill date was:

$$\hat{t}_{i+1} = t_i + d_i$$

- We defined a refill as **late** if the next observed fill occurred after a grace window:

$$t_{i+1} > t_i + d_i + g$$

  where $g = 7$ days.
- The **last fill in a chain was censored** because no subsequent refill is observed.

How we selected the best grouping

- We compared grouping levels by:
  - chain continuity / fragmentation,
  - number of usable refill events,
  - consistency of gap behaviour,
  - downstream model performance.
- **NDC7 gave the best balance**: less fragmentation than NDC11 and more stable refill histories for modelling.

2. Feature design

Main feature groups

- drug grouping
- current fill
- chain dynamics
- refill gap history
- patient utilization history
- payment and cost history
- same-day fill context
- calendar / time features

Dataset columns
132

Numeric
111

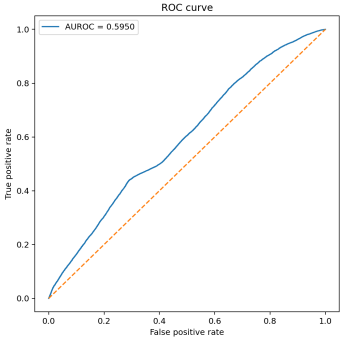
Categorical
4

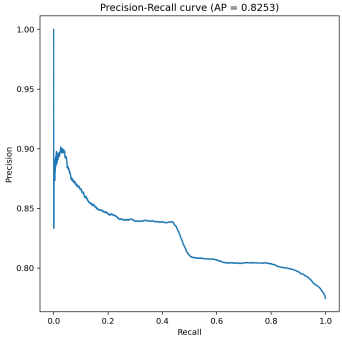
Model inputs
115

Categorical fields:
NDC11, NDC7, NDC8, PROD_SRVC_ID

3. Final XGBoost results

AUROC	PR-AUC	F1
0.6357	**0.8430**	**0.7496**

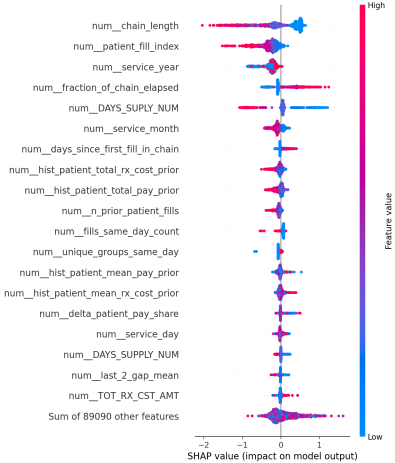




Validation setup: weighted training, best threshold = 0.44, validation prevalence = 0.7746. **ROC:** The model performs better than chance, but discrimination between late and non-late refill events remains moderate (AUROC = 0.6357).

the model shows stronger ranking performance, with PR-AUC = 0.8430, indicating useful prioritisation

4. Model comparison and interpretation



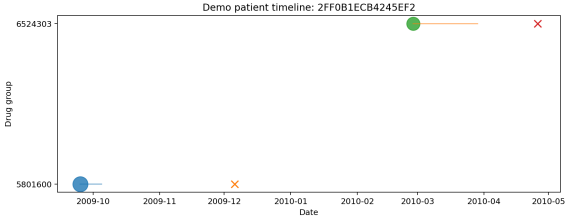
Models tested

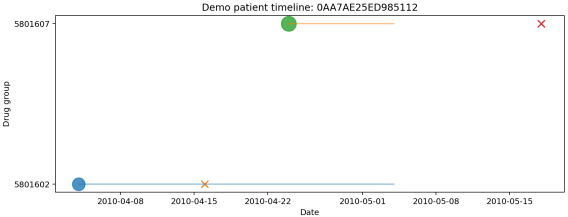
- LightGBM
- XGBoost
- FT-Transformer

Best FT-Transformer

- AUROC: **0.6095**
- PR-AUC: **0.8313**
- F1: **0.8791**
- Brier: **0.1674**

5. Patient demo





NDC7 gave the most stable refill chains and the clearest patient-level interpretation.

- Blue circle = an earlier fill in one drug group
- Green circle = a later fill in another drug group
- Horizontal line after each circle = expected coverage period from the days supply
- Orange X = the refill became late for that chain
- Red X = a later missed or late point for the second chain
- Y-axis = drug group ID
- X-axis = calendar date

Patient 1: Repeated long refill gaps across two drug groups.
Patient 2: Shorter refill history, but both chains still become late.

What the demo shows

- The model captures longitudinal refill behaviour rather than only one isolated transaction.
- High-risk predictions are often associated with:
  - short days supply,
  - long historical gaps,
  - advanced position within a refill chain,
  - increasing payment burden or utilization instability.

Take-away

- Choosing the right product grouping was essential.
- **NDC7** produced the most useful refill chains.
- Refill-history statistics were more informative than raw product identity alone.
- The patient demo makes the model outputs clinically and operationally interpretable.

Thank you